One-pot PCR amplicon library workflow

gene-region primers + barcode primers in the same tube → pool → Nanopore (Evoprep) · HuangLab LIMS stages 1 → 3a → 3b → 4a

Stage 1 — Specimen# / Culture#

Stage 3a — extraction + Qubit · MolWork#

Stage 3b — PCR + gel + Qubit · pcr_reaction row

Stage 4a — Nanopore experiment · Nanopore#

Specimen registered in LIMS
soil / environmental → Specimen# | fungal strain → Culture#

LIMS register
`/lims-register`
Stage 1 → Specimen# (soil / env.) or Culture# (strain)

Soil DNA extraction
SuperSoil protocol

Fungal DNA extraction
modified BOMB v9.0

DNA extraction

DNA template → MolWork#
soil crude extract (SuperSoil) | pure fungal extraction (BOMB v9)
MolWork# minted after quantification



Qubit Flex
quantify the extract (Stage 3a)

LIMS register
`/lims-register`
Stage 3a → MolWork# minted after quantification

Set up one-pot PCR

One reaction tube (per sample)

gene-region primers
(head-tailed, 0.1 µM each)

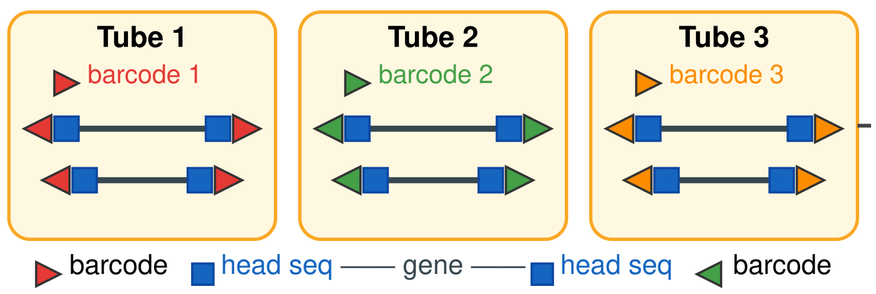
barcode primers
(barcode + head seq, 0.3 µM each)

50 µL: KAPA HiFi 2X + both primer sets + 3 µL template

One-pot PCR protocol
50 µL mix + 25/15-cycle program

One-pot PCR
phase 1: 25 cycles, anneal 60–66 °C (gene primers)
phase 2: 15 cycles, anneal 65 °C (barcode primers)

One-pot PCR products: one tube per sample, one barcode per tube





Gel check
confirm barcoded product

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

conc. → equal-molar volumes



Qubit Flex
quantify → 10 ng/µL (Stage 3b)

LIMS register
`/lims-register`
Stage 3b → pcr_reaction: conc., 2nd_pcr_ok, barcode

Pooled sample

Size selection 1X
bead purification protocol

Purification (bead clean-up)
PCR sample : bead = 1 : 1

Purified pooled sample

EvoPrep library prep
KAPA EvoPrep NBD114.96

Ready to be sequenced
library construction using Evoprep

LIMS register
`/lims-register`
Stage 4a → Nanopore#: experiment, samples, NanoAmp

Legend

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = LIMS stages; /lims-register at every stage; dashed = QC / info flow
single gel check: no intermediate 1st-PCR product to inspect